

NATIONAL INSTITUTE OF TECHNOLOGY AGARTALA



**DEPARTMENT- ELECTRONICS AND
INSTRUMENTATION ENGINEERING**

LAB NAME - NETWORK THEORY LABORATORY

TITLE - MOBILE INCOMING CALL INDICATOR

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INTRODUCTION

THIS PROJECT IS DESIGNED TO HELP ESCAPE FROM NUISANCE OF INCOMING CALL AT HOME, OFFICES AND DURING CONFERENCES. THIS CIRCUIT IS DESIGNED TO GIVE A VISUAL INDICATION IF PLACED NEAR A MOBILE PHONE EVEN IF THE RINGER IS DEACTIVATED.

WHEN A CALL IS COMING TO THE MOBILE PHONE, THE TRANSMITTER INSIDE IT BECOMES ACTIVATED. THE FREQUENCY OF THE TRANSMITTER IS AROUND 900MHZ. THE COIL PICKS UP THESE OSCILLATIONS BY INDUCTION AND FEEDS IT TO THE BASE OF THE TRANSISTOR. THIS MAKES THE TRANSISTOR ACTIVATED SINCE THE COLLECTOR OF THE TRANSISTOR IS CONNECTED TO THE PIN 2 OF 555, THE IC IS TRIGGERED TO MAKE THE LED CONNECTED AT ITS OUTPUT PIN TO BLINK. THE BLINKING OF THE LED IS THE INDICATION OF INCOMING CALL.

BLOCK DIAGRAM

• HARDWARE COMPONENTS

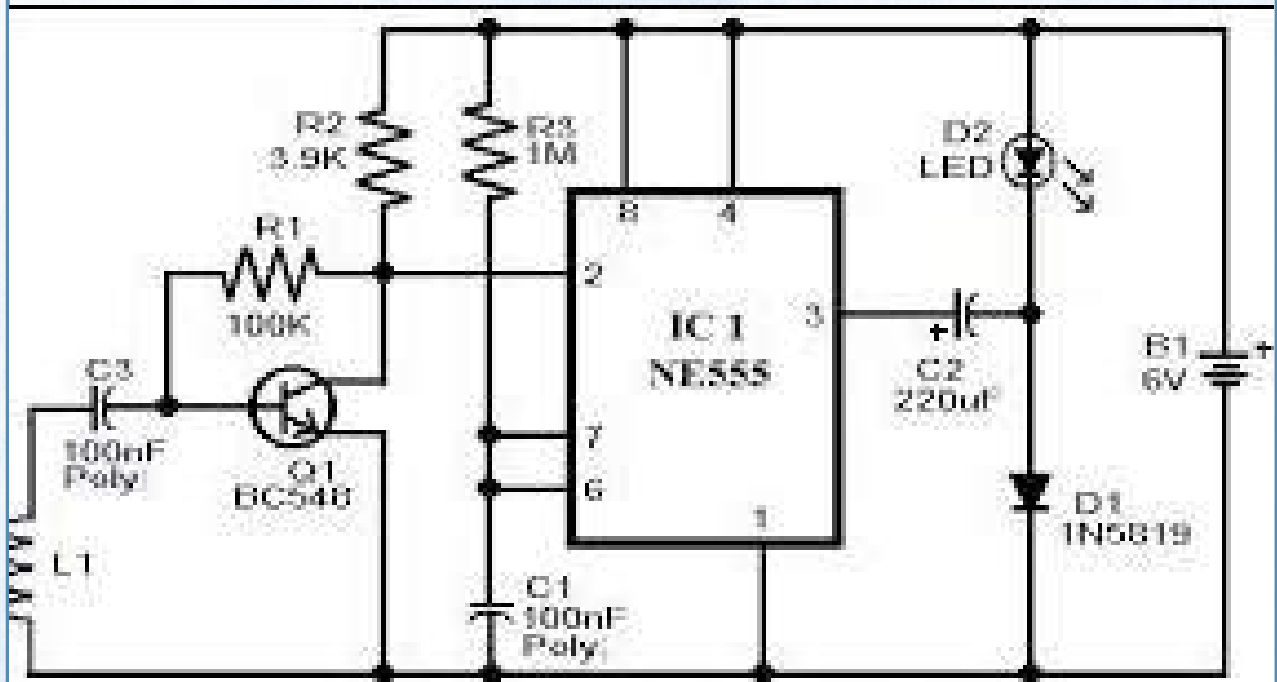
- IC 1: NE555(TIMER IC)
- RESISTORS: 100K,3.9K, 1M
- POLYESTER CAPACITOR: 100NF (63V)
- ELECTROLYTIC CAPACITOR: 220MICROF (25V)
- LED: 1A 40V SCHOTTKY-BARRIER DIODE
- TRANSISTOR: 100MILLIA 45V NPN
TRANSISTOR
- IC: 555 TIMER
- INDUCTOR: 10 MICRO HENRY

ABSTRACT

THIS CIRCUIT IS DESIGNED TO GIVE AN INDICATION FOR INCOMING CALLS IN A CELL PHONE. THIS ELECTRONIC PROJECT PROVES TO BE A RELIEF FROM THE NUISANCE CREATED DUE TO THE SUDDEN RINGING OF THE MOBILE.

THERE ARE MANY SITUATIONS WHERE WE CANNOT SWITCH OFF THE MOBILE NOR PUT IT IN SILENT MODE, YET A LOUD RING CAN PROVE TO BE VERY EMBARRASSING. THIS CIRCUIT PROVES TO BE A RELIEF IN SUCH SITUATIONS.

CIRCUIT DIAGRAM



Mobile incoming call indicator

BLOCK DIAGRAM



WORKING PRINCIPLE

1. RF Signal Emission by Mobile Phone:

- When a mobile phone is about to receive a call, it communicates with the nearest cell tower by emitting RF signals in the range of about 900 MHz to 1800 MHz (depending on the frequency band used by the mobile network).

2. Signal Detection by Antenna/Coil:

- The antenna or coil in the detector circuit picks up these RF signals from the mobile phone. The signal is typically very weak and requires amplification.

3. Tuning and Signal Filtering:

- The tuning circuit (usually an LC circuit) is set to resonate at the frequency range of the mobile phone's RF signals. This helps in filtering out other frequencies and focusing on the mobile phone's signal.

4. Rectification:

- The AC signal detected by the antenna is then passed through a diode, which acts as a rectifier, converting it into a pulsating DC signal.

5. Signal Amplification:

- This weak DC signal is fed into the base of a transistor, which amplifies it. The transistor essentially acts as a switch that is turned on by the presence of the RF signal.

6. Indicator Activation:

- Once the signal is amplified, it is strong enough to power an LED or a buzzer. The LED will light up, or the buzzer will sound, indicating that the mobile phone is receiving a call

7. Output:

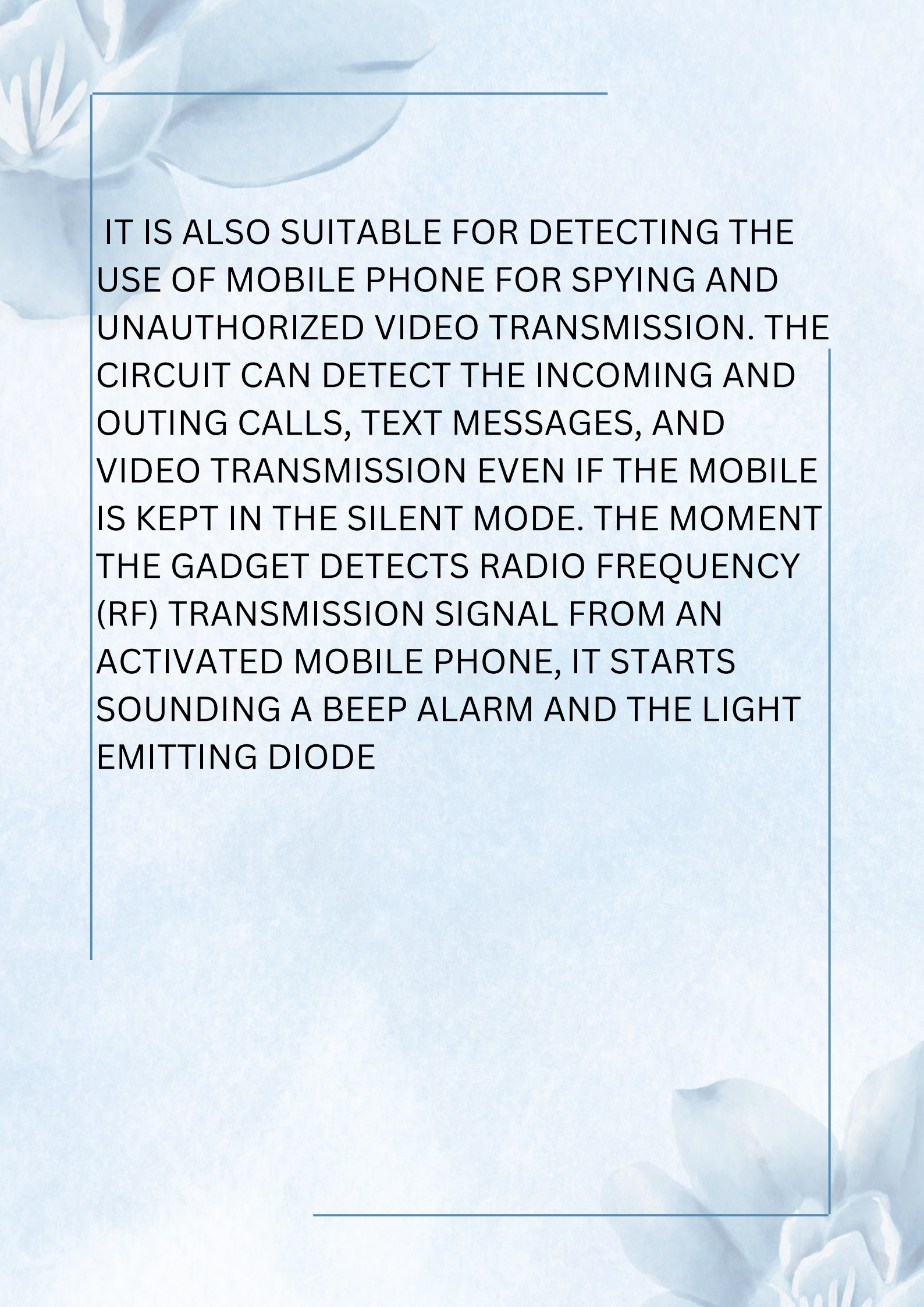
- The user is alerted by the visual or auditory signal, even if the mobile phone is set to silent or vibrate mode.

FUTURE SCOPE

- THE SYSTEM WAS BUILT USING A 555 TIMER IC, COIL, PICKUP ANTENNA AND SO MANY OTHER ELECTRONICS COMPONENTS.
- THE MOST IMPORTANT PART OF THE CIRCUIT IS A 130-150 TURN COIL, WHICH ACTS AS A PICK-UP ANTENNA FOR A SIGNAL THAT IS EMITTED BY YOUR MOBILE PHONE WHENEVER IT HAS AN INCOMING CALL. THE SIGNAL INDUCED IN THIS COIL/INDUCTOR IS AMPLIFIED THROUGH A TRANSISTOR, WHICH IS THEN USED TO DRIVE A MONOSTABLE 555 TIMER.
- THE OUTPUT VOLTAGE OF THE IC IS USED TO DRIVE A ULTRA-BRIGHT LED.

CONCLUSION

- IN CONCLUSION THIS CIRCUIT CAN BE USED TO ESCAPE FROM THE NUISANCE OF MOBILE PHONE RINGS WHEN YOU ARE AT HOME. THIS CIRCUIT WILL GIVE A VISUAL INDICATION IF PLACED NEAR A MOBILE PHONE EVEN IF THE RINGER IS DEACTIVATED. THIS WORK INVOLVES THE DESIGN AND DEVELOPMENT OF A DIGITAL SIGNAL DETECTOR WHICH IS CAPABLE OF DETECTING INCOMING AND OUTGOING SIGNALS FROM MOBILE PHONES.
- THE PRESENCE OF AN ACTIVATED MOBILE PHONE CAN BE DETECTED BY THIS HANDY, POCKET-SIZE MOBILE SIGNAL DETECTOR FROM A DISTANCE OF ONE AND A HALF METERS, WHICH COULD BE USED IN PREVENTING THE USE OF MOBILE PHONES IN EXAMINATION HALLS, CONFIDENTIAL ROOMS ETC.



IT IS ALSO SUITABLE FOR DETECTING THE USE OF MOBILE PHONE FOR SPYING AND UNAUTHORIZED VIDEO TRANSMISSION. THE CIRCUIT CAN DETECT THE INCOMING AND OUTING CALLS, TEXT MESSAGES, AND VIDEO TRANSMISSION EVEN IF THE MOBILE IS KEPT IN THE SILENT MODE. THE MOMENT THE GADGET DETECTS RADIO FREQUENCY (RF) TRANSMISSION SIGNAL FROM AN ACTIVATED MOBILE PHONE, IT STARTS SOUNDING A BEEP ALARM AND THE LIGHT EMITTING DIODE

REFERENCES

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7. robu.in
8. Book : - Charles K. Alexander, Matthew N. O. Sadiku,, “Fundamentals of Electric Circuits”,